DM IMCQ 2
Spring 2026

Anatomy, Histology, Biochemistry

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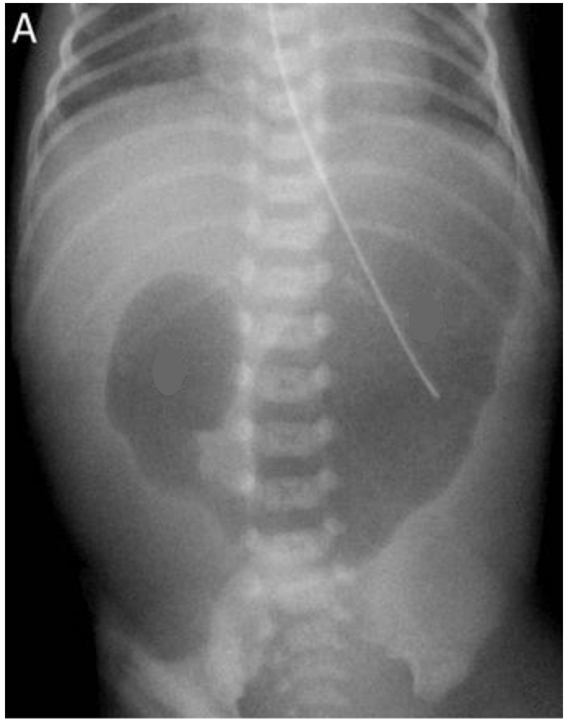
Anatomy 3 Questions

A 12-hour-old full-term male newborn is seen by the physician because of a non-bilious vomiting. The pregnancy was complicated by polyhydramnios. The patient was delivered by vaginal delivery and did not pass meconium. Physical examination shows mild jaundice and mild upper abdominal distension. Plain radiograph of the patient is shown. During laparoscopic surgery, the cause of the obstruction is found to be glandular tissue encircling the duodenum. The obstructed bowel is cut and re-anastomosed. Which of the following best describes the cause of bowel obstruction in the patient?



A. Hypertrophy of smooth muscle
B. Extrinsic compression of gut tube
C. Complete failure of recanalization
D. Dorsal deviation of dividing septum
E. Failed migration of neural crest cells

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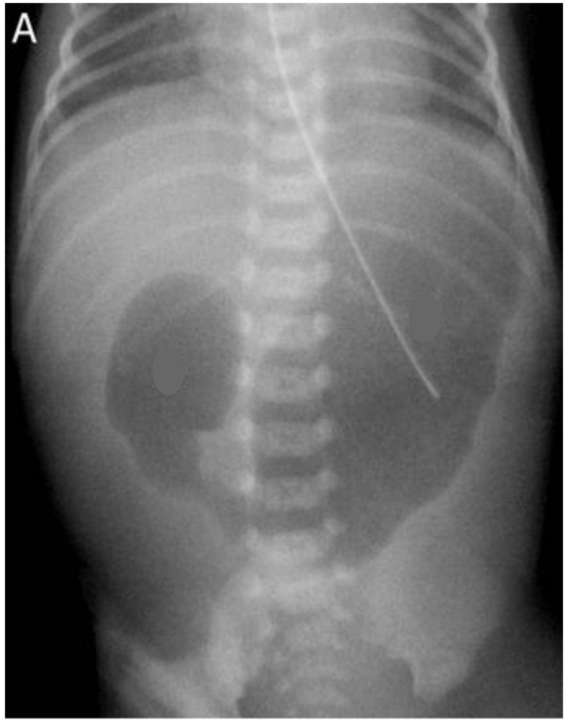

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SOM.MK.1.BPM2.3.DM.1.ANAT.1104
Describe the embryologic mechanism responsible for formation of annular pancreas and its clinical significance

A 73-year-old woman comes to the emergency department because of constant epigastric pain for the past 3 days. She also reports dark colored urine, pale stool and jaundice for the past 2 days. She had a cholecystectomy 1 year ago. Blood pressure is 126/80 mm Hg, pulse is 90/min, respirations are 16/min and temperature is 38.9°C (102°F). Physical examination shows yellowing of the skin and sclera, and soft abdomen with mild epigastric and right upper quadrant tenderness. Laboratory results show elevated bilirubin and liver enzymes. CT-scan of the abdomen shows numerous stones in the biliary tree with severe dilation extending up to just proximal to the ductal opening in the duodenum and no pancreatic ductal dilation. The stones are most likely located in which of the following parts of the biliary tree?

A. **Cystic duct**
B. **Gallbladder**
C. **Ampulla of vater**
D. **Common Bile duct**
E. **Right hepatic duct**

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B. Gallbladder
C. Ampulla of vater
✓ **D. Common Bile duct**
E. Right hepatic duct

SOM.MK.1.BMP2.3.DM.1.ANAT.1168

Describe the anatomy of the biliary tree.

A 56-year-old man comes to the emergency department because of a 2-day history of severe epigastric pain, nausea and vomiting. He also reports a 13.6-kg (30-lb) weight loss, and early satiety during the past 3 weeks. He has noted a mass growing in the supraclavicular region for the past 8 months. He has an extensive history of GERD, and kidney disease. Physical examination shows a non-tender, 6-cm soft tissue mass superior to the left clavicle. Colonoscopy shows a 6-cm mass in the proximal transverse colon. Pathology reports of the masses from both locations show moderately differentiated adenocarcinoma, confirming primary colon cancer. To which of the following lymphatic structures would the cancer cells have most likely drained initially?

A. Celiac nodes
B. Thoracic Duct
C. Cisterna Chyli
D. Supraclavicular nodes
E. Superior Mesenteric nodes

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SOM.MK.1.BMP2.3.DM.1.ANAT.1151
Describe the location, arterial supply, and the venous and lymphatic drainage of the rectum and anal canal.

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Histology 6 Questions

1. A 51-year-old man is brought to the emergency department because of a 3-hour history of severe abdominal pain radiating to the right shoulder. Physical examination shows jaundice, and tenderness in the right upper quadrant of his abdomen. Laboratory studies show elevated levels of conjugated bilirubin. Abdominal ultrasonography shows 'stones' (lithiasis) within the lumen of the common bile duct and gall bladder. The hormone cholecystokinin (CCK) is known to stimulate contraction of this organ. Which of the following best identifies the origin of this hormone?

A. Mucus cells
B. Parietal cells
C. Microfold cells
D. Enteroendocrine cells
E. Enterocytes

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Differentiate between the cells of the gastric mucosa based on their histological structures, functions and localization. SOM.MK.I.BPM2.3.DM.1.HCB.1215
Distinguish between the cells of the small intestinal mucosa based on location, function and microscopic featuresSOM.MK.I.BPM2.3.DM.1.HCB.1235

2. A 55-year-old woman is brought to the emergency department because of a 1-day history of mental confusion and rigors, and a 3-day history of worsening abdominal pain and jaundice. She has a history of gallstones. Her blood pressure is 101/75 mm Hg, pulse is 110/min, respirations are 24/min and temperature is 40^{o}C (104^{o}F). Physical examination shows a febrile man with jaundice and right upper quadrant tenderness. Laboratory studies show elevated liver enzymes and bilirubin levels, and leukocytosis. Abdominal imaging confirms an acute cholangitis. This condition is characterized by acute inflammation and infection of the biliary ductal system, coupled with the obstruction of bile flow. Which of the following best characterizes the affected ductal system?

A. Receives secretory components from an endothelial lined structure called the central vein
B. In some parts are lined by specialized epithelial cells called cholangiocytes
C. Carries 25% of blood received by the liver
D. The muscular layer in its walls are stimulated by secretin
E. The initial part leaves the secretory acini as centroacinar cells

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Describe the microscopic structure of the intrahepatic biliary tree.SOM.MK.I.BPM2.3.DM.1.HCB.1299

3. A 33-year-old woman is brought to the emergency department because of a 1-day history of fainting spells and a 2-month history of generalized weakness and dizziness. Her blood pressure is 103/79 mm Hg, and pulse is 112/min. Physical examination shows pale mucous membranes. A digital rectal examination shows dark tarry stool on the examination finger. Tests of stool for occult blood are positive. Endoscopy shows a large ulcer in the fundus of the stomach. Which of the following cell types is most likely to be functionally impaired in this condition?

A. Chief
B. Parietal
C. Enteroendocrine
D. Surface mucous
E. Paneth

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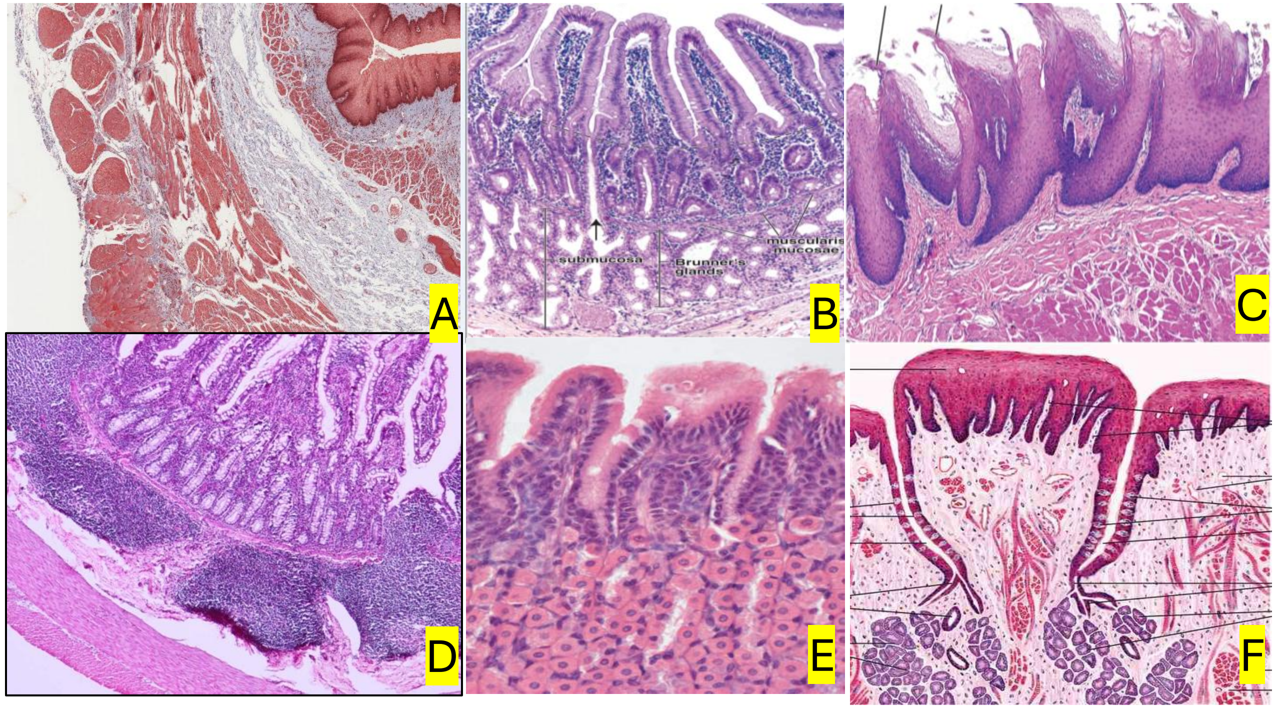
Distinguish between the cells of the small intestinal mucosa based on location, function and microscopic featuresSOM.MK.I.BPM2.3.DM.1.HCB.1235

4. A 25-year-old man comes to the physician because of a 2-week history of a black tongue. He is a chronic smoker. His blood pressure is 120/70 mm Hg, and pulse is 70/min. Physical examination shows a black discoloration of the tongue. Examination of biopsy specimens taken from random areas of his tongue show destruction of the circumvallate papillae. Which of the following labeled photomicrographs best corresponds to the affected structure?

A. A
B. B
C. C
D. D
E. E
F. F



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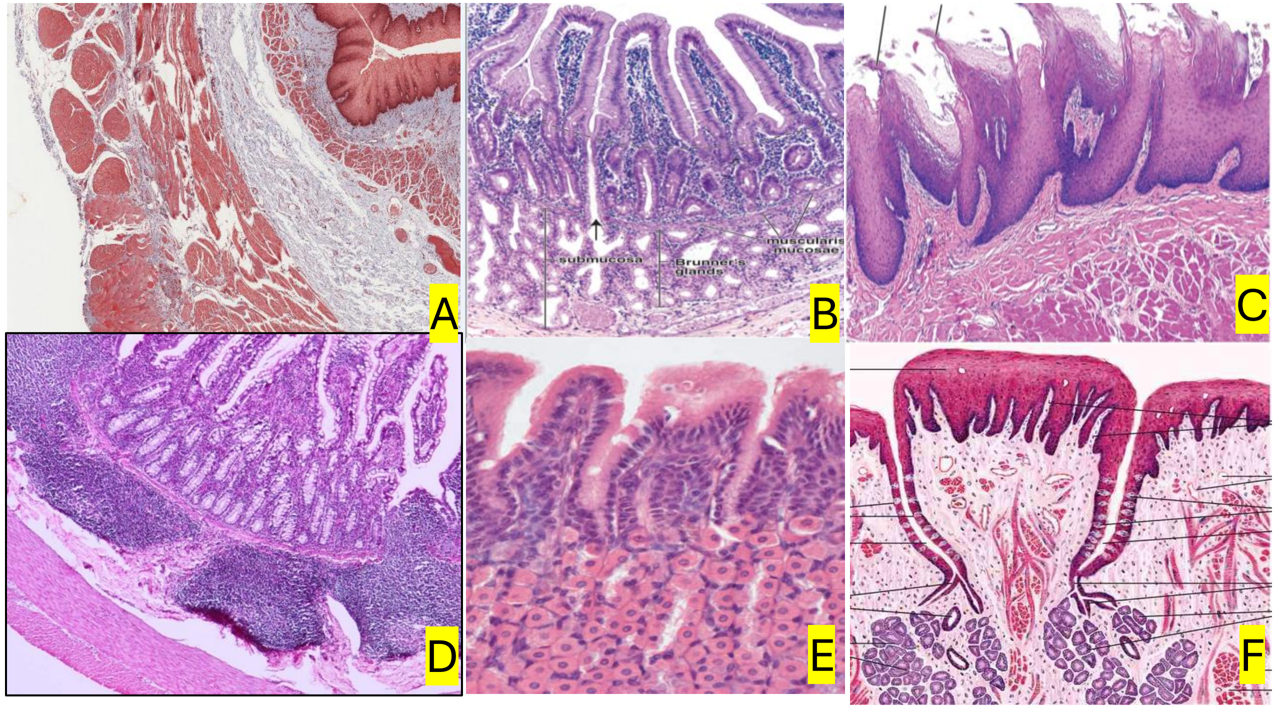

Identify the microscopic structure of the lingual papillae. SOM.MK.I.BPM2.3.DM.1.HCB.1195

5. A 50-year-old man comes to the physician because of a 3-month history of blood in the stool, pain on defecation and a sensation of fullness in the anus. His vital signs are all within normal limits. Physical examination shows a bleeding mass in the anal canal. Microscopic examination of biopsy specimens of the mass shows that it originates from epithelial cells within the anal canal. Which of the following best characterizes the structure in which the mass was found?

A. Peyer's patches in the lamina propria
B. Lower 1/3 is lined by stratified squamous epithelium
C. Presence of rugae
D. Brunner's glands in the submucosa
E. Well-developed plicae circularis

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Describe the histological appearance of the structure (including the 3 zones) of the anal canal.SOM.MK.I.BPM2.3.DM.1.HCB.1264

6. A 51-year-old woman is brought to the emergency department because of a 1-month history of worsening abdominal pains, constipation, intermittent blood in the stool, and a 6.8-kg (15-lb) weight loss. Physical examination shows pale oral mucosa and conjunctivae. A colonoscopy shows a mass in the descending colon that bleeds easily. Examination of biopsy specimens from the mass shows cells with multiple mitoses and loss of cellular architecture. Which of the following is the most likely gross feature in this affected region of the gastrointestinal tract?

A. Haustra
B. Rugae
C. Villi
D. Lacteals
E. Plicae

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Describe the microscopic differences typically seen in each of the parts of the large intestine.SOM.MK.I.BPM2.3.DM.1.HCB.1252

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Biochemistry 7 Questions

An 8-month-old boy is brought to the emergency department by his mother because of a generalized seizure episode. She says that this is his first episode of seizures. He has a 3-day history of fever and vomiting and has been unable to eat or drink anything since. Physical examination shows a lethargic infant who is only responsive to painful stimuli. His fingerstick blood glucose concentration is low. Urine dipstick is negative for glucose and ketones. Symptoms are relieved following glucose infusion. Which of the following hepatic enzymes is most likely lacking an allosteric activator during fasting in this patient?

A. Glycogen phosphorylase
B. Pyruvate carboxylase
C. Phosphoenolpyruvate carboxykinase (PEPCK)
D. Alanine aminotransferase
E. Citrate synthase
F. Pentose dehydrogenase complex

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SOM. MK.I.BPM2.3.DM.2.BCHM.1202 Compare and contrast systemic beta-oxidation defects and myopathic beta oxidation defects. Identify causes for each.

A. Glycogen phosphorylase
✓ B. Pyruvate carboxylase
C. Phosphoenolpyruvate carboxykinase (PEPCK)
D. Alanine aminotransferase
E. Citrate synthase
F. Pentose dehydrogenase complex

A 10-year-old boy is brought to the physician by his parents because of a 2-month history of recurrent painful muscle cramping during school sports. Various laboratory tests are performed. The forearm ischemic exercise test shows an increase of lactate in the blood during exercise. Laboratory studies show low serum carnitine. Urinalysis shows the presence of carnitine in his urine. A muscle biopsy specimen shows the presence of lipid droplets. Which of the following processes is most likely impaired in this patient?

A. Mitochondrial fatty acid oxidation in liver
B. Transport of fatty acids into mitochondria
C. Peroxisomal oxidation of C-26 fatty acids
D. Muscle glycogenolysis
E. Liver glycogenolysis
F. Oxidation of branched chain fatty acids

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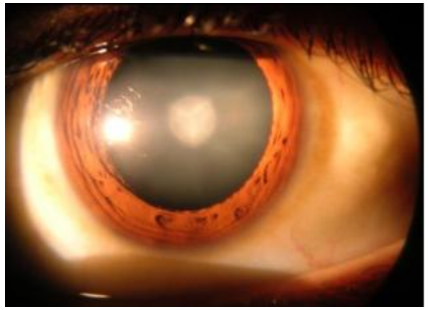
SOM. MK.I.BPM2.3.DM.2.BCHM.1202; Compare and contrast systemic beta-oxidation defects and myopathic beta oxidation defects. Identify causes for each.

A 1-month-old girl is brought to the physician by her parents because of progressive yellowish discoloration of her skin, poor feeding and frequent vomiting. She was born at 38-weeks' gestation. She has been exclusively breastfed but vomits after feeds. Physical examination shows ~~a~~ jaundice. Abdominal examination shows hepatomegaly. Eye examination shows the findings in the photograph. Increased activity of which of the following enzymes best explains the eye changes in this patient?



A. Galactose 1-phosphate uridyl transferase
B. Galactokinase
C. Fructokinase
D. Aldolase B
E. Aldose reductase

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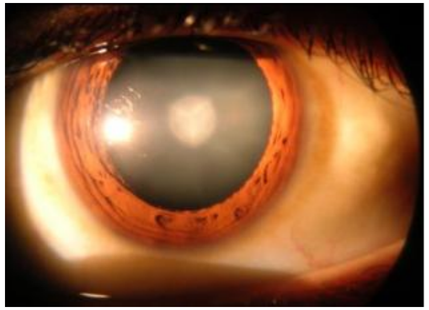

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E. Aldose reductase

SOM. MK.I.BPM2.3.DM.2.BCHM.1169 Describe the polyol pathway, its function and its importance in diabetic patients.

A 37-year-old man of Afro-Caribbean descent comes to the physician because of a 2-day history of dark urine and yellowing of the eyes. He recently started taking primaquine for malaria prophylaxis before an overseas work trip. Physical examination shows scleral icterus. Laboratory studies show anemia and the presence of bite cells and Heinz bodies on peripheral blood smear. The patient is diagnosed with a disorder that impairs erythrocyte NADPH production. Which of the following enzymes most likely provides an alternative source of cytoplasmic NADPH for fatty acid synthesis in other tissues?

A. Glucose 6-phosphate dehydrogenase
B. Acetyl CoA carboxylase
C. Mitochondrial malate dehydrogenase
D. Cytoplasmic malate dehydrogenase
E. Citrate synthase

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D. Cytoplasmic malate dehydrogenase
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SOM. MK.I.BPM2.3.DM.2.BCHM.1186 Identify the sources of NADPH for fatty acid synthesis.

A 3-day-old girl is brought to the emergency department because of yellowish discoloration of the skin and poor feeding. She has had repeated episodes of nonbilious, nonprojectile, vomiting after being breastfed. Physical examination shows scleral icterus and bilateral opacity of lenses. Abdominal examination shows a palpable liver edge 2-cm below the right costal margin. Laboratory studies show elevated alanine aminotransferase (ALT) and aspartate aminotransferase, and urine is positive for reducing substances. A blood culture is positive for *Escherichia coli*. She is started on fluid resuscitation and intravenous antibiotics. Which of the following is the best next step in management of this patient?

A. Continuous infusion of uncooked starch
B. Sucrose-free diet
C. Soy-based formula
D. Recombinant acid α-glucosidase (acid maltase)
E. No additional intervention is needed
F. Thiamine supplements

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SOM. MK.I.BPM2.3.DM.2.BCHM.1171; Compare and contrast laboratory findings, enzyme deficiency, clinical manifestations and management of Classical and non-classical galactosemia. SOM. MK.I.BPM2.3.DM.2.BCHM.1172; Differentiate benign fructosuria, galactosemia, lactose intolerance and hereditary fructose intolerance based on clinical features and laboratory findings. Review the dietary management of these disorders.

A 7-year-old boy is brought to the emergency department by his parents because of a 1-hour history of continuous, unremitting seizures. He is stabilized and treated for status epilepticus. He has a history of language disorder, intellectual developmental delay and failure to thrive. Physical examination shows muscle weakness and facial dysmorphism. A homozygous variant in the *ACACA* gene, which encodes acetyl-CoA carboxylase-1, is detected by whole-exome sequencing and confirmed by Next-Gen sequencing. It is observed that there is impaired regulation of an enzyme due to less product formation by the deficient enzyme. Regulation of which of the following enzymes is most likely impaired in this patient?

A. Fatty acyl CoA synthetase (thiokinase)
B. Carnitine palmitoyl transferase-I (CPT-1)
C. ATP citrate lyase
D. Hormone-sensitive lipase
E. Citrate synthase
F. Glycogen synthase

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E. Citrate synthase
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SOM. MK.I.BPM2.3.DM.2.BCHM.1195 Describe the carnitine shuttle mechanism.

A 28-year-old woman comes to the physician for a routine health maintenance examination. She is advised to come the following week to obtain blood samples for estimation of fasting serum glucose concentration 10 hours after her last meal (dinner). Liver metabolism during fasting shows the degradation of a molecule that inhibits hepatic fructose 1,6-bisphosphatase. Which of the following enzymes most likely degrades this molecule?

A. Pyruvate carboxylase
B. Phosphorylated bifunctional enzyme
C. Phosphoenolpyruvate carboxykinase
D. Pyruvate dehydrogenase (PDH) kinase
E. Phosphorylated pyruvate kinase
F. Phosphofructokinase-1 (PFK-1)

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SOM. MK.I.BPM2.2.DM.3.BCHM.1140 Describe the hepatic bifunctional enzyme and distinguish the role of glucagon and insulin on gluconeogenesis.

A. Pyruvate carboxylase
B. Phosphorylated bifunctional enzyme
C. Phosphoenolpyruvate carboxykinase
D. Pyruvate dehydrogenase (PDH) kinase
E. Phosphorylated pyruvate kinase
F. Phosphofructokinase-1 (PFK-1)